Here are some helpful YouTube links and online resources for BCA-504: Numerical Methods:

UNIT-I: Roots of Equations

1. **YouTube** - *Bisection Method for Solving Equations*
   https://www.youtube.com/watch?v=5X8FFmjwFSY
2. **YouTube** - *False Position Method for Roots of Equations*
   https://www.youtube.com/watch?v=4zRYQdZG6do
3. **YouTube** - *Newton's Raphson Method*
   https://www.youtube.com/watch?v=fmReiw2nTmA
4. **YouTube** - *Rate of Convergence of Newton's Method*
   https://www.youtube.com/watch?v=1C1M9OHh0xE
5. **GeeksforGeeks** - *Methods for Finding Roots of Equations*
   https://www.geeksforgeeks.org/numerical-methods-bisection-method/

UNIT-II: Interpolation and Extrapolation

1. **YouTube** - *Finite Differences and Newton's Forward & Backward Differences*
   https://www.youtube.com/watch?v=Skh2TQyDiP0
2. **YouTube** - *Lagrange's Interpolation Formula*
   https://www.youtube.com/watch?v=6XfDL_TpKZ0
3. **YouTube** - *Newton's Divided Difference Formula*
   https://www.youtube.com/watch?v=3IVjY3hvc9I
4. **YouTube** - *Gauss's Interpolation Formula*
   https://www.youtube.com/watch?v=QzixwI-7rS8
5. **GeeksforGeeks** - *Interpolation Methods: Lagrange, Newton, Gauss*
   https://www.geeksforgeeks.org/interpolation-using-lagranges-formula/

UNIT-III: Numerical Differentiation and Integration

1. **YouTube** - *Numerical Differentiation (Finite Differences)*
   https://www.youtube.com/watch?v=J6bmRlJzZJw
2. **YouTube** - *Trapezoidal Rule for Numerical Integration*
   https://www.youtube.com/watch?v=mi4vUHOtjyI
3. **YouTube** - *Simpson's One-Third Rule for Numerical Integration*
   https://www.youtube.com/watch?v=jhr-ytIuvtQ
4. **YouTube** - *Simpson's Three-Eighths Rule*
   https://www.youtube.com/watch?v=FYV4eEuHZNE
5. **GeeksforGeeks** - *Numerical Integration Methods*
   https://www.geeksforgeeks.org/numerical-integration-using-trapezoidal-rule/

UNIT-IV: Solution of Linear Equations

1. **YouTube** - *Gauss's Elimination Method for Solving Linear Equations*
   https://www.youtube.com/watch?v=8uQSQFj2jb4
2. **YouTube** - *Gauss's Seidel Iterative Method*
   https://www.youtube.com/watch?v=2zqG4rPf9vQ
3. **GeeksforGeeks** - *Gauss-Seidel Method for Solving Linear Equations*
   https://www.geeksforgeeks.org/gauss-seidel-method/

UNIT-V: Solution of Differential Equations

1. **YouTube** - *Euler's Method for Solving Differential Equations*
   https://www.youtube.com/watch?v=KMS95F0dC-I
2. **YouTube** - *Picard's Method for Solving Differential Equations*
   https://www.youtube.com/watch?v=-yFVYFlWfDA
3. **YouTube** - *Fourth-Order Runge-Kutta Method*
   https://www.youtube.com/watch?v=jItN0PHt0Xk
4. **GeeksforGeeks** - *Runge-Kutta Methods*
   https://www.geeksforgeeks.org/runge-kutta-methods/

Additional Resources:

1. **TutorialsPoint** - *Numerical Methods Overview*
   https://www.tutorialspoint.com/numerical_methods/index.htm
2. **Wolfram MathWorld** - *Numerical Methods Reference*
   https://mathworld.wolfram.com/topics/NumericalMethods.html

These resources will help you understand the core concepts and methods for **Numerical Methods**, including solving equations, interpolation, differentiation, integration, and differential equations.